

Artificial Intelligence in Education: Research Gap Analysis and Structured Abstract

Part A: Research Gap Analysis

Title of the Paper [Reference]	Methodology	Dataset	Limitation	Identified Research Gap
Prediction of Student Performance Using Machine Learning Techniques [1]	Data cleaning, missing-value handling, and Recursive Feature Elimination (RFE) for feature selection; multiple ML classifiers compared, with Linear Regression performing best	Real-world dataset collected from multiple educational institutions (demographic, academic, and behavioral attributes)	Relies mainly on static, structured academic/demographic data; does not use real-time behavioral or engagement signals, and offers no explainability of predictions	Need for models that fuse real-time learning-engagement data with explainable AI to support early, interpretable intervention
Prediction of Student's Performance by Modelling Small Dataset Size [2]	Multiple supervised ML classifiers trained and evaluated at progressively increasing dataset sizes	Small student cohort dataset from a single university course	Small, single-institution dataset limits generalizability; performance is highly sensitive to dataset size	Lack of validated, generalizable models tested across large, diverse, multi-institutional datasets

Table 1: Research Gap Analysis of Existing Work on AI-Based Student Performance Prediction

Part B: Research Abstract

Title: An Explainable AI Framework for Early Prediction of Student Performance Using Multi-modal Academic and Behavioral Data

Predicting student academic performance is essential for enabling timely, targeted interventions in higher education. Existing machine learning approaches typically rely on static academic and demographic records analyzed with classical classifiers, or are validated only on small, single-institution datasets. Consequently, these models struggle to generalize across diverse student populations, overlook real-time behavioral and engagement signals, and rarely provide interpretable justifications for their predictions – a critical requirement for educators to trust and act upon algorithmic recommendations. This research addresses these gaps by proposing a hybrid, explainable machine learning framework that fuses structured academic records with behavioral and engagement data (such as assignment submission patterns, login frequency, and participation metrics) collected from learning management systems. The proposed methodology involves multimodal data preprocessing, feature fusion, and training ensemble classifiers (Random Forest and Gradient Boosting), followed by the application of SHAP (SHapley Additive exPlanations) to interpret model predictions and identify the key drivers of academic risk. The framework will be evaluated on a multi-institutional dataset to test generalizability across varied learning environments. It is expected that the proposed model will achieve higher predictive accuracy than single-source academic models while producing transparent, human-interpretable explanations for at-risk classifications. The significance of this work lies in bridging the gap between predictive accuracy and explainability in educational AI, enabling instructors to identify struggling students earlier and take informed, personalized corrective action, thereby contributing to improved learning outcomes and more equitable educational support systems.

Keywords: Artificial Intelligence in Education, Student Performance Prediction, Explainable AI, Educational Data Mining, Machine Learning

Part C: References

References

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